

Graphic

①

$$\max \quad 5x_1 + 7x_2 = f(x_1, x_2)$$

s.t.

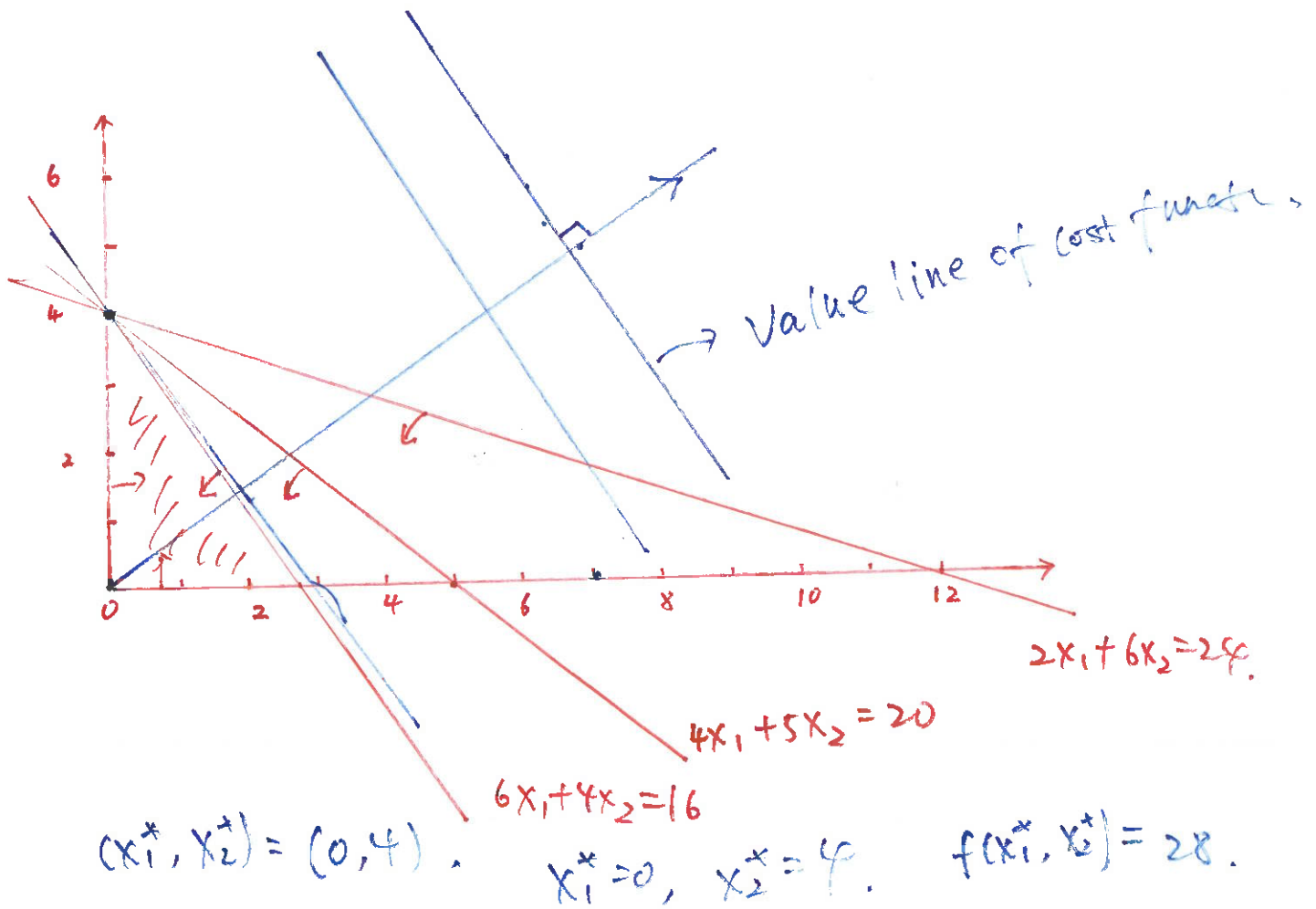
$$4x_1 + 5x_2 \leq 20$$

$$2x_1 + 6x_2 \leq 24$$

$$6x_1 + 4x_2 \leq 16$$

$$x_1, x_2 \geq 0$$

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$



(2)

Simplex

$$\max \quad 5x_1 + 7x_2 = Z \quad Z - 5x_1 - 7x_2 = 0$$

s.t.

$$4x_1 + 5x_2 \leq 20 \quad 4x_1 + 5x_2 + s_1 = 20$$

$$2x_1 + 6x_2 \leq 24 \quad 2x_1 + 6x_2 + s_2 = 24$$

$$6x_1 + 4x_2 \leq 16 \quad 6x_1 + 4x_2 + s_3 = 16$$

$$x_1, x_2 \geq 0$$

↑
b.

	Z	x_1	x_2	s_1	s_2	s_3	b
R_1	1	-5	(-7)	0	0	0	0
R_2	0	4	(5)	1	0	0	20. $20/5=4$
R_3	0	2	6	0	1	0	24. $24/6=4$
R_4	0	6	4	0	0	1	16. $16/4=4$

$$x_1=0, x_2=0, s_1=20, s_2=24, s_3=16, Z=0.$$

$$R_2 \leftarrow R_2/5$$

$$R_1 \leftarrow R_1 + 7R_2$$

$$R_2$$

$$R_3 \leftarrow R_3 - 6R_2$$

$$R_4 \leftarrow R_4 - 4R_2$$

	Z	x_1	(x_2)	s_1	(s_2)	(s_3)	b
	1	0.6	0	1.4	0	0	28
	0	0.8	1	0.2	0	0	4
	0	-2.8	0	-1.2	1	0	0
	0	2.8	0	-0.8	0	1	0

$$x_1=0, x_2=4, s_1=20, s_2=0, s_3=0.$$

③

Branch - and - Bound

$$\max 7x_1 + 5x_2$$

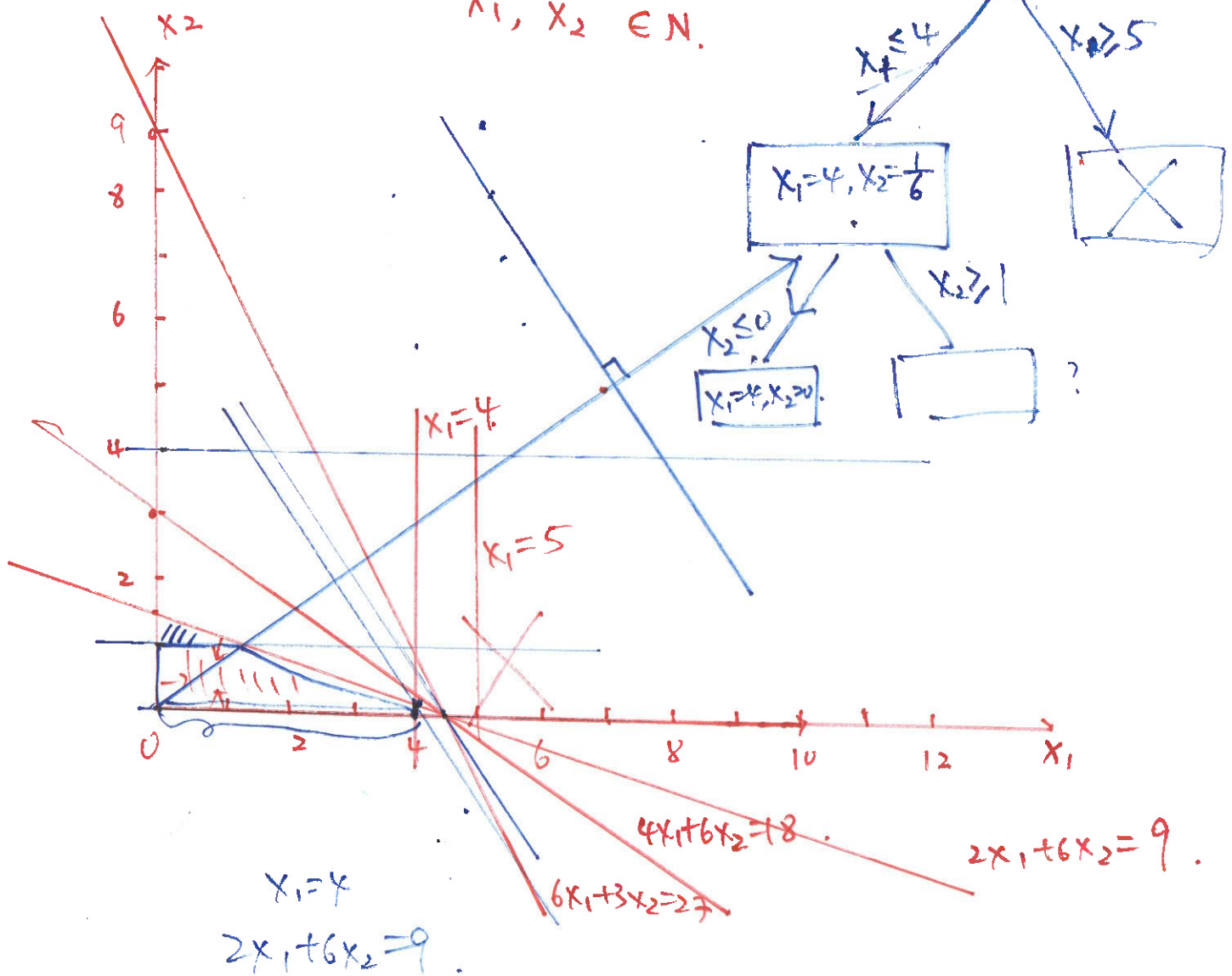
s.t.

$$4x_1 + 6x_2 \leq 18$$

$$2x_1 + 6x_2 \leq 9$$

$$6x_1 + 3x_2 \leq 27$$

$$x_1, x_2 \in \mathbb{N}.$$



(4)

Graphic

max

$$200x + 550y$$

s.t.

$$100x + 250y \leq 5000$$

$$x + y \leq 50$$

$$x, y \geq 0$$

5

Simplex

$$\max \quad 200x + 550y$$

$$\text{s.t.} \quad 100x + 250y \leq 1000$$

$$x + y \leq 50$$

$$x, y \geq 0$$